
IM0802 – Responsible Artificial Intelligence

Assignment 2: “Paper Title Goes Here”

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Abstract

1 Introduction 350

Israel, which allocates 8.8% of its GDP to defense [1], invests heavily in the development of autonomous weapon systems (AWS), with their Prime Minister announcing a \$100 billion investment in homegrown arms production last December [2]. Israel’s most well-known homegrown AWS is the Iron Dome air defense system, which intercepts incoming projectiles in real time. Building on this foundation in artificial intelligence (AI)-assisted defense technology, Israel has also expanded into offensive technology, such as the ‘Habsora’ targeting system.

Habsora (‘The Gospel’) is an AI-targeting system used by the Israeli Defense Force (IDF), developed primarily by Unit 8200 within the Military Intelligence Directorate. First deployed during the May 2021 Operation ‘Guardian of the Walls’, it generates target recommendations using a multimodal intelligence pipeline that encompasses signal intelligence, telephone metadata, satellite and drone imagery, social graph analysis, and life-pattern modelling [3, 4]. The system is an ensemble of multiple specialised algorithms fused together [4]. During the 2023-2024 Gaza conflict, Habsora was reported to generate over one hundred target recommendations per day, which in previously would require approximately twelve months of manual analysis [3, 5].

Habsora’s AI capability is the probabilistic classification and prioritisation of locations and individuals as military objectives. It assigns confidence scores derived from large-scale pattern matching across a shared intelligence pool, rather than positive identification in the traditional intelligence analysis [6]. Military commanders authorise each recommendation, yet the review window documented for each target was approximately only twenty seconds [3].

This creates a stakeholder dynamic with both ethical and legal significance. Its output influencing both IDF commands, who operate under high cognitive load and time pressure; Palestinian civilians in the target environment bearing the lethal consequences and the broader international community, which depends on International Humanitarian Law (IHL) compliance for the legitimacy of armed conflict. Steen et al. [7] offer a framework for this tension. They propose a two-axis model of Meaningful Human Control (MHC) which distinguishes between high automation and genuine human control. Both require adequate information and real freedom of action. The central argument of this report is that Habsora structurally undermines the second condition, and that this gap carries cascading implications across the technical, sociotechnical, and governance layers identified by Verdues et al. [8] in their Comprehensive Human Oversight Framework (CHOF).

2 Ethical Issue Identification 750

Verdues et al. [8]’s CHOF was developed to assess accountability for autonomous systems whose internals are not directly inspectable. It divides oversight into a 3×3 matrix, before, during and

	Before	During	After
Technical	Partial Built; no XAI embedded	Absent No reasoning chain shown	Absent No performance audit
Sociotechnical	Partial No bias mitigation training	Absent 20-sec rubber stamp dynamic	Absent No incident review process
Governance	Absent No EIA or legal pre-approval	Absent No IHL compliance check	Absent No audit or accountability

Figure 1: The CHOF framework for structuring ethical analysis across governance, socio-technical, and technical layers. Two yellow cells indicate where Habsora can be given partial compliance. The red cells indicate absence.

after deployment across a technical, socio-technical and governance mapping. The resulting matrix surfaces gaps which remain hidden when ‘human oversight’ would be treated as a single requirement. Applying this framework to Habsora’s implementation we get Figure 1.

2.1 Technical Layer

Habsora has been described as an ensemble of deep learning models processing multimodal data [3, 4]. Such architectures are inherently opaque, Arrieta et al. [9] characterise this as the accuracy/interpretability trade-off, where deep ensembles maximize predictive performance at the cost of transparency. While post-hoc explainability methods that could be applied to such systems exist, reports on Habsora’s use indicate that no such methods are applied at the operators point of decision.

This opacity directly links to legal consequences. Schmitt and Thurnher [6] establish that IHL’s distinction principle requires positive identification of a target as being a military objective. In conjunction with the proportionality assessment being understood by the user. Without any interpretability, the legal review is effectively empty.

The above failures map directly onto the highest-ranked principles identified by Khan et al. [10]. Both **P-01** transparency and **P-03** accountability, further compounded by their challenges: lack of technical understanding among decision-makers (**CH-06**) and the absence of binding legal frameworks (**CH09**). While this might not be a failure of intent, it is a structural property of opaque architectures. Combined with lacking operator-facing interfaces, this creates a system that cannot provide the context for understanding.

2.2 Socio-technical

The Value Sensitive Design (VSD) framework addresses indirect stakeholders, those affected without directly using the system, such as civilians in the target environment. Palestinian civilians were not involved in its design, and their values were not elicited to ground the system’s targeting logic. They remain non-consenting parties who face the consequences of the system’s use without recourse. There is a large asymmetry in consequences between those operating the system and those subjected to it.

For IDF commanders, the conditions necessary for genuine human oversight are structurally not present. Steen et al. [7] argue that meaningful human control requires both sufficient information and meaningful freedom of action. Habsora undermines both: commanders receive recommendations without transparency into the underlying reasoning, and the short decision windows of the system leave almost no room for critical evaluation. Miller [11] shows that from social science, without understanding what went into a decision, the explainee cannot meaningfully interrogate it. Combined with the large amount of targets the system generates, this reduces what would normally require ex-

tended deliberation and proportionality assessments to a button press. Former IDF commanders have described this process as ‘rubber-stamping’ [12], where those with authority are merely approving the recommendations made by an AWS. This confirms the presence of automation bias, which is a phenomena where operators rely too much on AI outputs under time pressure [13].

The consequence is that IDF commanders formally bear moral and legal responsibility for strikes they cannot meaningfully evaluate, while the system’s opacity diffuses accountability in practice. Targeting decisions that once demanded careful human judgement are now effectively delegated to an algorithm. This prevents the human control which International Humanitarian Law demands, and that Steen et al. [7] identify as a precondition for attributing responsibility.

2.3 Governance

Article 36 of Protocol I to the Geneva Conventions requires countries to perform legal reviews of new weapons systems to determine whether their use would be prohibited under international law [14]. Israel is one of the few countries that have not ratified Protocol I [15], meaning that from a governance perspective, they are not obligated to consider the legal consequences of systems like Habsora. UNESCO maintains that delegating life-or-death decisions to an AI system is itself forbidden [16]. Many countries comply with the UNESCO ruling, but since there is no public information available confirming whether Israel also complies, we can deduce that no such accountability frameworks exist in the case of Habsora.

Once the weapon systems are deployed, establishing accountability becomes difficult. Under international humanitarian law, commanders are legally responsible for how weapons are used, even when autonomous functions are involved [17]. They are judged on what they knew at the time of the decision [18]. In practice, responsibility is spread across developers, the algorithm, the receiving IDF officer, and the approving commander [16]. The Lavender system, the human-targeting extension of Habsora, has a documented 10% error tolerance in targeting recommendations, which might imply that Habsora itself also has a high error tolerance. This margin makes civilian casualties inevitable. In the midst of war, real-time monitoring is not easy to pull off, which compounds the issue.

After deployment, we find that review processes are largely absent. The system’s opacity makes it difficult to determine whether casualties resulted from an AI recommendation or a human override. Palestinian civilians, who the most directly affected stakeholders, were never consulted.

3 Responsible Solution Design 900

3.1 Technical

One technical failure identified in 2, is that operators receive target recommendations without a reasoning chain. The direct remedy would be a transparency-by-design philosophy, only given that these models would trigger the accuracy/interpretability trade-off [9], this adoption would be unlikely. The realistic path would therefore become post-hoc explainability, integrated into the operators user interface. With this, as noted by Arrieta et al. [9]’s audience-centred definition: explainability is always relative to a given audience, the implementation should be calibrated to the field-operator, not for a Machine Learning engineer debugging the model weights.

We propose two post-hoc interpretability methods, SHapley Additive exPlanations (SHAP) [19] and Gradient-weighted Class Activation Mapping (Grad-CAM). SHAP assigns each input feature a contribution score for a prediction. For example a targeting recommendation for a building can be driven for 47% by the targets movement patterns, for 30% by their phone’s metadata, and for 23% by using drone footage showing targets entering it. Additionally, SHAP is model-agnostic and is therefore applicable to Habsora’s ensemble architecture, without accessing the models internals. Combined with Grad-CAM, which highlights images to show which parts contributed to the predictions (ref image), creates an evidential basis of each recommendation.

However, raw feature attribution is necessary but not sufficient. Miller [11] identifies four properties of human explanation: they’re contrastive (why X rather than Y), selective (not exhaustive), causal rather than probabilistic, and involve a social process between the explainer and explainee. SHAP’s raw output does not satisfy any of these properties and must therefore be expanded upon by providing the operator with a contrastive explanation: not only recommending the ‘best’ target, but also presenting

the nearest non-target alternative alongside its scores. This allows the operator to evaluate 'why this target rather than no strike', critically assess the recommendation, and override it if necessary.

Liao et al. [20]'s XAI 'QuestionBank' operationalises this further. Their taxonomy identifies questions which users need answered to make informed decisions: 'why', 'why not' and 'what if'. These map directly into IHL requirements: *Why* supports distinction, *Why not* supports proportionality, and *What if* enables the precautionary assessment. The design of the interface should have these as mandatory fields, not optional features. Morley et al. [21] note that post-hoc explainability tools tend to only be used at the testing stage of the ML-pipeline. Our recommendation insists that this should also be implemented in the operational deployment, not only during the systems validation.

3.2 Socio-technical

As noted earlier, even if the technical layer would provide perfect explanations, the operational tempo would still undermine meaningful use. Steen et al. [7] define two conditions of moral responsibility: adequate information and meaningful freedom of action. Our proposed mechanism is a mandatory minimum deliberation time, with a hard interface lock. The operator cannot approve any strike before the review period has elapsed, ensuring that the deliberation standard is met by design, and preventing premature confirmation.

Wasilow and Thorpe [16]'s LOAC compliance framework (Category 3) specifies what this deliberation standard must contain. Instead of a single button press approving the recommendation, the workflow should decompose this into confirmations aligned with IHL's core principles: *distinction* (is this a valid military target?), *proportionality* (is the anticipated military advantage proportionate to the expected civilian harm?), and *precaution* (have all measures to minimise civilian harm been made?). For each question a separate confirmation must be given, which also serves as the legal standard as identified by Schmitt and Thurnher [6], and prevents a multi-step legal judgement into a single button press.

Moreover, Parasuraman and Manzey [13] demonstrate that automation complacency increases with system reliability and task monotony. The interface must actively interrupt this cycle by requiring engagements checks in the explanation layer: an operator should not be able to click through the IHL checklist without first reviewing the contrastive explanations and feature weights suggested before. This creates a forced interaction with the XAI output, counteracting that the explanation becomes a goal to be bypassed by the operator.

Miller [11] reinforces the case for active engagement by specifying the types of questions an explanation must answer to be actionable: why a target was selected, why an alternative was not, and what would change under different inputs.

The head-up display (HUD) for the operator could contain information, such as confidence, reliability and uncertainty, that makes the use of Habsora in that particular instance explainable. For example, the system can show a percentage likelihood that the strike will hit the right target. Amershi et al. [22]'s human-AI interaction guidelines, specify that systems must calibrate displayed confidence to actual reliability and communicate uncertainty explicitly. Each of these principles directly addresses the trust miscalibration that underlies automation bias.

Friedman et al. [23]'s VSD methodology requires that indirect stakeholders' values are embedded in the system's design. Civilian presence in the target environment cannot remain pure metadata. Instead, value-laden indicators such as civilian density or protected-site proximity should be surfaced in the operator interface as a structural requirement. Considering the troublesome relationship between the Palestinians and the Israelis, such a process would likely require intermediary agents to facilitate the process, like the United Nations or another impartial international body.

3.3 Governance

Stahl et al. [24]'s three-level mitigation typology provides a useful organising frame for the governance solutions we propose. Firstly, because it maps onto the temporal dimension of the CHOF matrix, making it clear where each intervention fits in the system lifecycle. And secondly, because it emphasises that policy-level strategies, guidance mechanisms, and corporate governance measures must be implemented together to effectively address the identified gaps in governance.

3.3.1 Pre-deployment

A mandatory pre-deployment ethical impact assessment (EIA) should be grounded in UNESCO’s recommendation against delegating life-or-death decisions to AI [25]. This would ensure that human rights implications are systematically reviewed before deployment. Voluntary review processes, like Canada’s legal assessment protocol [16], would be insufficient. Binding policy-level requirements are necessary to prevent the kind of gaps in governance we see in Israel’s non-ratification of Protocol I to the Geneva Conventions. This gap can be categorised using Khan et al. [10]’s CH-09: the absence of institutionalised ethical review mechanisms in AI deployment.

3.3.2 During deployment

Glass Box compliance norms should be followed during deployment: documented input/output thresholds for the system’s recommendations, mandatory escalation criteria for when human operators must intervene, and post-strike logs recording which recommendation was authorised and on what basis. According to Verdiesen et al. [8], this should not remain an optional design choice, but rather an operational legal requirement.

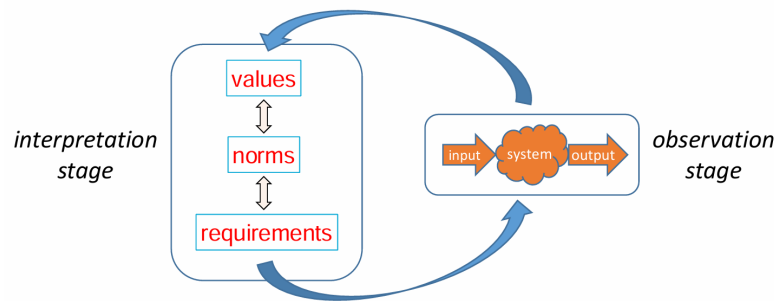


Figure 2: Glass Box compliance norms for real-time IHL adherence.

3.3.3 Post-deployment

An independent auditor should have access to the Glass Box logs to verify compliance with International Humanitarian Law and identify any violations. This would address the responsibility diffusion problem by replacing it with a traceable chain of authorisation, as Wasilow & Thorpe suggest [16]. This would restore accountability and the possibility of redress for affected parties, addressing the gaps identified in Khan et al. [10]’s P-01 and P-03 regarding the absence of accountability mechanisms and the erosion of human agency in AI systems.

None of these levels of governance intervention is sufficient on their own. The CHOF matrix makes this explicit by showing how the absence of governance measures at each stage of the system lifecycle creates gaps that can be exploited.

4 Discussion and Conclusions 400

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Appendix

Abbreviations

AI	Artificial Intelligence
AWS	Autonomous Weapon Systems
CH-06	Challenge 06: Lack of technical understanding among decision-makers [10]
CH-09	Challenge 09: Absence of binding legal frameworks [10]
CHOF	Comprehensive Human Oversight Framework [8]
EIA	Ethical Impact Assessment
HCI	Human-Computer Interaction
IDF	Israel Defense Forces
IHL	International Humanitarian Law
LAWS	Lethal Autonomous Weapon Systems
LIME	Local Interpretable Model-agnostic Explanations
MHC	Meaningful Human Control [7]
ML	Machine Learning
P-01	Principle 01: Transparency [10]
P-03	Principle 03: Accountability [10]
SHAP	SHapley Additive exPlanations
UX	User Experience
VSD	Value Sensitive Design [23, 26]
XAI	Explainable Artificial Intelligence

Glossary

Term	Definition
Term	Definition in the format of a sentence
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